AI Tutor

A unified learning platform for university students

Generative Ai And Ai Platform Engineering - Course Project

Problem Statement

Students need reliable, personalised study support that is available at any hour and stays true to what their course teaches.

The need	AI Tutor's response
Explanations that match their level and course	**Topic lessons with examples and practice questions**
Help understanding diagrams, lectures and spoken answers	**Image, video and audio analysis**
Answers drawn only from their own course material	**Retrieval-based answers with source references**
Personal feedback on their marks, even without internet	**Fine-tuned local advisor and offline mode**

Objectives

1. Generate lessons, quizzes and viva feedback as structured output
2. Generate text and complete sentences with controllable sampling
3. Answer course questions strictly from uploaded material
4. Work online and offline, with automatic fallback
5. Give personalised progress reports in one web interface

Four Project Modules

1

Multimodal Tutor

CO1 · CO4

Lessons, diagrams, viva and video quizzes

2

Text Generation

CO2

Models that generate and complete text

3

Course Q&A (RAG)

CO3

Answers grounded in course PDFs

4

Advisor and Platform

CO5 · CO6

Fine-tuned advisor and React app

Module 1: Multimodal Tutor

Covers CO1 and CO4

Definition A Gemini-powered API that turns text, images, audio and video into structured study support, using carefully designed and versioned prompts.

What it does

- Topic lessons with examples and practice
- Diagram and image explanation
- Viva answer scoring from audio
- Quiz generation from short videos
- Validated JSON with retry and fallback

Tools

FastAPI

Pydantic

Google GenAI SDK

Gemini 3.8 Flash

Module 2: Text Generation

Covers CO2

Definition Trains and tests small GPT-style models that learn from text and complete sentences, showing how a generative model turns tokens into fluent continuations.

What it does

- Subword tokenization with token-count comparison
- Small GPT-style models trained from scratch
- Sentence completion from custom text prompts
- Sampling control: temperature, top-k, top-p
- Token and cost estimate for every feature

Tools

PyTorch

Hugging Face Transformers

Datasets

Module 3: Course Q&A with RAG

Covers CO3

Definition Retrieves passages from uploaded course PDFs and answers strictly from them, so every response stays tied to the syllabus. Runs with Gemini online or Ollama offline.

What it does

- PDF upload, chunking and embedding
- Vectors in Chroma, metadata in MongoDB
- Optional course and CO filter
- Top-5 retrieval with cited sources
- Retrieval quality: Hit Rate@k and MRR

Tools

Sentence-transformers

ChromaDB

MongoDB

Ollama

Module 4: Advisor and Platform

Covers CO5 and CO6

Definition Fine-tunes a compact local model to produce accurate student progress reports, and brings every feature together in one React web application with production-ready operations.

What it does

- Qwen2.5-1.5B fine-tuned with QLoRA
- Accurate rank, strengths and advice as JSON
- React + Vite front end for all features
- Tracing, guardrails and a kill switch
- Docker, CI checks and AWS deployment

Tools

Ollama and PEFT

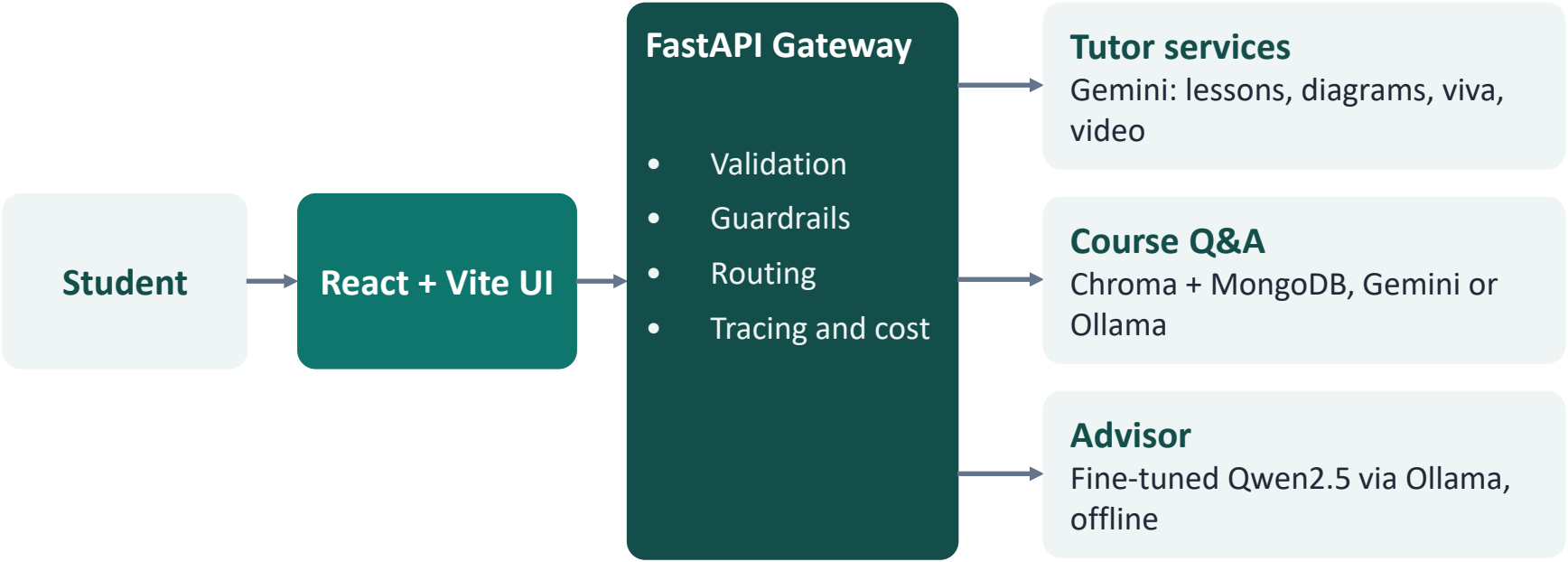
React + Vite

Docker

GitHub Actions

Terraform

System Architecture



Every response is validated, traced and returned as structured JSON

Project Flow

1

Ask

Student picks a feature in the web app

2

Validate

Inputs checked and unsafe requests blocked

3

Route

Sent to tutor, course Q&A or advisor

4

Generate

Answer built by the model or retrieved passages

5

Return

Structured JSON shown, traced and logged